
BE 502: Design and Analysis of Bioalgorithms

Course Code	BE 502
Course Name	Design and Analysis of Bioalgorithms
L-T-P-C	2-0-2-3
Prerequisites	IC136, IC152, IC260
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	10th Senate (Jun 2016)

Course Contents

- **Introduction**

Background on genomics and proteomics, DNA-RNA gene, protein structures, Importance of Bio-algorithms, Bioinformatics algorithms and examples of algorithms designs, algorithm complexity and speed, Bio-signal processing algorithms and design considerations [4 Lectures]

- **Combinatorial pattern matching algorithms for genomes**

Hash Tables, Exact matching, Suffix Trees, Keyword Trees, Heuristic similarity search methods, approximate pattern matching, sequence similarity search, sequence alignment, BLAST, and motif finding [9 Lectures]

- **Graph-based and Clustering algorithms**

Graph terminologies and some basic algorithms, shortest superstring-based DNA sequencing, hybridization-based DNA sequencing, graphs for peptide sequencing, K-means clustering and Hierarchical clustering for gene expressions, graph-based clustering, phylogenetic tree reconstruction. [9 Lectures]

- **Bio-signal processing**

Pre-processing considerations in bio-signals (for EEG and FMRI), signal decomposition methods, graph-based and clustering algorithms for EEG and FMRI, feature extraction and their uses in diagnosis of diseases. [6 Lectures]

Text Books

1. N.C. Jones and P.A. Pevzner, **An introduction to bioinformatics algorithms**, MIT Press, 2004.
2. K. Najarian, R. Splinter, **Biomedical Signal and Image Processing**, 2nd Edition, CRC Press, 2012.

References

1. T. Cormen, C. Leiserson, R. Rivest, C. Stein, **Introduction to algorithms**, 3rd Edition, MIT Press, 2009.